ENEX 254 — Electromagnetics

Formula Sheet

Chapter-wise · formulas & reference tables

6 chapters · 27 topics · 206 items

Contents

Chapter 1: Introduction — Coordinate Systems & Vector Fields

2 topics · 24 items

1.1 Reference Formulas for Chapter 1 *(Reference)*

(G) Quadrant rule for ϕ — the most common source of error

Quadrant	Signs	Correction	Range
I	$x > 0,\ y > 0$	$\phi = \tan^{-1} \lVert y/x \rVert$	0°–90°
II	$x < 0,\ y > 0$	$\phi = 180° - \tan^{-1} \lVert y/x \rVert$	90°–180°
III	$x < 0,\ y < 0$	$\phi = 180° + \tan^{-1} \lVert y/x \rVert$	180°–270°
IV	$x > 0,\ y < 0$	$\phi = 360° - \tan^{-1} \lVert y/x \rVert$	270°–360°

(H) Differential elements — line, surface and volume

System	Line $d\mathbf{l}$	Surface $d\mathbf{S}$ (constant-coordinate faces)	Volume dv
Cartesian	$dx\,\hat{a}_x + dy\,\hat{a}_y + dz\,\hat{a}_z$	$dy\,dz\,\hat{a}_x$; $dx\,dz\,\hat{a}_y$; $dx\,dy\,\hat{a}_z$	$dx\,dy\,dz$
Cylindrical	$d\rho\,\hat{a}_\rho + \rho\,d\phi\,\hat{a}_\phi + dz\,\hat{a}_z$	$\rho\,d\phi\,dz\,\hat{a}_\rho$; $d\rho\,dz\,\hat{a}_\phi$; $\rho\,d\rho\,d\phi\,\hat{a}_z$	$\rho\,d\rho\,d\phi\,dz$
Spherical	$dr\,\hat{a}_r + r\,d\theta\,\hat{a}_\theta + r\sin\theta\,d\phi\,\hat{a}_\phi$	$r^2\sin\theta\,d\theta\,d\phi\,\hat{a}_r$; $r\sin\theta\,dr\,d\phi\,\hat{a}_\theta$; $r\,dr\,d\theta\,\hat{a}_\phi$	$r^2\sin\theta\,dr\,d\theta\,d\phi$

1.1.1

$$\rho = \sqrt{x^2 + y^2}, \qquad \phi = \tan^{-1}\frac{y}{x}, \qquad z = z$$

1.1.2

$$x = \rho\cos\phi, \qquad y = \rho\sin\phi, \qquad z = z$$

1.1.3

$$A_\rho = A_x\cos\phi + A_y\sin\phi$$

1.1.4

$$A_\phi = -A_x\sin\phi + A_y\cos\phi$$

1.1.5

$$A_z = A_z$$

1.1.6

$$A_x = A_\rho\cos\phi - A_\phi\sin\phi$$

1.1.7

$$A_y = A_\rho\sin\phi + A_\phi\cos\phi$$

1.1.8

$$A_z = A_z$$

1.1.9

$$r = \sqrt{x^2 + y^2 + z^2}, \qquad \theta = \cos^{-1}\frac{z}{r} = \tan^{-1}\frac{\sqrt{x^2 + y^2}}{z}, \qquad \phi = \tan^{-1}\frac{y}{x}$$

1.1.10

$$x = r\sin\theta\cos\phi, \qquad y = r\sin\theta\sin\phi, \qquad z = r\cos\theta$$

1.1.11

$$A_r = A_x\sin\theta\cos\phi + A_y\sin\theta\sin\phi + A_z\cos\theta$$

1.1.12

$$A_\theta = A_x \cos\theta \cos\phi + A_y \cos\theta \sin\phi - A_z \sin\theta$$

1.1.13

$$A_\phi = -A_x \sin\phi + A_y \cos\phi$$

1.1.14

$$\hat{a}_\rho = \sin\theta\,\hat{a}_r + \cos\theta\,\hat{a}_\theta$$

1.1.15

$$\hat{a}_z = \cos\theta\,\hat{a}_r - \sin\theta\,\hat{a}_\theta$$

1.1.16

$$\hat{a}_\phi = \hat{a}_\phi, \qquad z = r\cos\theta$$

1.2 Coordinate Transformation & Vector/Scalar Fields

1.2.1

$$\mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z$$

1.2.2

$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

1.2.3

$$\mathbf{a}_A = \frac{\mathbf{A}}{|\mathbf{A}|}$$

1.2.4

$$\mathbf{A}\cdot\mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta = A_xB_x + A_yB_y + A_zB_z$$

1.2.5

$$\mathbf{A}\times\mathbf{B} = |\mathbf{A}||\mathbf{B}|\sin\theta\,\mathbf{a}_n = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

1.2.6

$$|\mathbf{A}| = \sqrt{3^2 + (-4)^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$$

Chapter 2: Electric Field

8 topics · 65 items

2.1 Reference Formulas for Chapter 2 *(Reference)*

(C) Standard charge configurations (results derived in this chapter)

Configuration	**E**	Notes
Point charge Q	$\dfrac{Q}{4\pi\varepsilon_0 R^2}\,\hat{a}_R$	R = distance charge → point
Infinite line charge ρ_L	$\dfrac{\rho_L}{2\pi\varepsilon_0 \rho}\,\hat{a}_\rho$	ρ = perpendicular distance from the line
Infinite sheet charge ρ_S	$\dfrac{\rho_S}{2\varepsilon_0}\,\hat{a}_N$	independent of distance; $\hat{a}_N$ points from the sheet toward the field point

(H) Differential elements — essential for the integrals

System	Line $d\mathbf{l}$	Surface $d\mathbf{S}$ (constant-coordinate faces)	Volume dv
Cartesian	$dx\,\hat{a}_x + dy\,\hat{a}_y + dz\,\hat{a}_z$	$dy\,dz\,\hat{a}_x$; $dx\,dz\,\hat{a}_y$; $dx\,dy\,\hat{a}_z$	$dx\,dy\,dz$
Cylindrical	$d\rho\,\hat{a}_\rho + \rho\,d\phi\,\hat{a}_\phi + dz\,\hat{a}_z$	$\rho\,d\phi\,dz\,\hat{a}_\rho$; $d\rho\,dz\,\hat{a}_\phi$; $\rho\,d\rho\,d\phi\,\hat{a}_z$	$\rho\,d\rho\,d\phi\,dz$
Spherical	$dr\,\hat{a}_r + r\,d\theta\,\hat{a}_\theta + r\sin\theta\,d\phi\,\hat{a}_\phi$	$r^2\sin\theta\,d\theta\,d\phi\,\hat{a}_r$; $r\sin\theta\,dr\,d\phi\,\hat{a}_\theta$; $r\,dr\,d\theta\,\hat{a}_\phi$	$r^2\sin\theta\,dr\,d\theta\,d\phi$

2.1.1

$$\varepsilon_0 = 8.854 \times 10^{-12}\ \text{F/m}, \qquad \frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \text{m/F}, \qquad \frac{1}{2\pi\varepsilon_0} = 1.7975 \times 10^{10}\ \text{m/F}$$

2.1.2

$$\mathbf{F}_{12} = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R^2}\,\hat{a}_{R_{12}}, \qquad \mathbf{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\,\hat{a}_R, \qquad \mathbf{D} = \varepsilon_0\mathbf{E} = \frac{Q}{4\pi R^2}\,\hat{a}_R$$

2.1.3

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = Q_{\text{enc}} \quad \text{(integral form)}, \qquad \nabla\cdot\mathbf{D} = \rho_v \quad \text{(point form)}$$

2.1.4

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_v (\nabla\cdot\mathbf{D})\,dv$$

2.1.5

$$\int_0^{\pi/2} \sin^2\phi\,d\phi = \int_0^{\pi/2} \cos^2\phi\,d\phi = \frac{\pi}{4}, \qquad \int \sin^2\phi\,d\phi = \frac{\phi}{2} - \frac{\sin 2\phi}{4}$$

2.1.6

$$\int \sin\theta\cos\theta\,d\theta = \frac{1}{2}\sin^2\theta \quad \text{(or equivalently } \tfrac{1}{2} \text{ over } 0 \to \pi/2\text{)}, \qquad \int \sin 2\theta\,d\theta = -\tfrac{1}{2}\cos 2\theta$$

2.2 Coulomb's Law, Field Intensity, Flux Density & Gauss's Law

2.2.1

$$\mathbf{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{Q_1 Q_2}{R^2}\mathbf{a}_R = 9 \times 10^9 \frac{Q_1 Q_2}{R^2}\mathbf{a}_R$$

2.2.2

$$F = \frac{1}{4\pi\varepsilon_0}\frac{|Q_1 Q_2|}{R^2}$$

2.2.3

$$F = 9 \times 10^9 \frac{(1 \times 10^{-6})(2 \times 10^{-6})}{(0.5)^2} = 9 \times 10^9 \frac{2 \times 10^{-12}}{0.25} = 0.072\,\text{N}$$

2.3 Current Density, Continuity Equation & Relaxation Time

(U) Relaxation time

Material	n with T	μ with T	σ with T	Temp. coeff.
Metal	constant	decreases	decreases	positive
Semiconductor	increases (exp.)	decreases	increases	negative

2.3.1

$$I = \int_S \mathbf{J}\cdot d\mathbf{S}, \qquad \mathbf{J} = \sigma\mathbf{E} \quad \text{(conduction, point form of Ohm's law)}, \qquad \mathbf{J} = \rho_v\mathbf{u} \quad \text{(convection)}$$

2.3.2

$$\oint_S \mathbf{J}\cdot d\mathbf{S} = -\frac{dQ_i}{dt} \quad \text{(integral form)}, \qquad \nabla\cdot\mathbf{J} = -\frac{\partial \rho_v}{\partial t} \quad \text{(point form)}$$

2.3.3

$$\rho_v = \rho_{v0}\, e^{-t/\tau}, \qquad \tau = \frac{\varepsilon}{\sigma}\ \text{(seconds)}$$

2.3.4

$$I = \int_S \mathbf{J}\cdot d\mathbf{S}, \qquad \mathbf{J} = \sigma\mathbf{E} = \rho_v \mathbf{v}$$

2.3.5

$$\nabla\cdot\mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

2.3.6

$$\tau = \frac{\varepsilon}{\sigma}$$

2.4 Divergence & Divergence Theorem

2.4.1

$$\nabla\cdot\mathbf{A} = \lim_{\Delta v\to 0} \frac{1}{\Delta v}\oint_S \mathbf{A}\cdot d\mathbf{S}$$

2.4.2

$$\int_v (\nabla\cdot\mathbf{A})\, dv = \oint_S \mathbf{A}\cdot d\mathbf{S}$$

2.4.3

$$\nabla\cdot\mathbf{A} = 2+1+1 = 4$$

2.4.4

$$\text{Flux} = \int_v 4\, dv = 4\times(1) = 4$$

2.5 Electric Boundary Conditions

2.5.1

$$\oint \mathbf{E}\cdot d\mathbf{L} = E_{t1}\Delta w - E_{t2}\Delta w + (\text{terms } \propto \Delta h) = 0$$

2.5.2

$$E_{t1}\Delta w = E_{t2}\Delta w \quad\Longrightarrow\quad E_{t1} = E_{t2}$$

2.5.3

$$\frac{D_{t1}}{\varepsilon_1} = \frac{D_{t2}}{\varepsilon_2}$$

2.5.4

$$\oint \mathbf{D}\cdot d\mathbf{S} = D_{n1}\Delta S - D_{n2}\Delta S + (\text{side walls} \to 0 \text{ as } \Delta h \to 0) = \rho_s \Delta S$$

2.5.5

$$D_{n1} - D_{n2} = \rho_s$$

2.5.6

$$D_{n1} = D_{n2} \quad\Longrightarrow\quad \varepsilon_1 E_{n1} = \varepsilon_2 E_{n2}$$

2.5.7

$$\tan\theta_1 = \frac{E_{t1}}{E_{n1}}, \qquad \tan\theta_2 = \frac{E_{t2}}{E_{n2}}$$

2.5.8

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{E_{n2}}{E_{n1}} = \frac{\varepsilon_1}{\varepsilon_2} \quad \Longrightarrow \quad \frac{\tan\theta_1}{\tan\theta_2} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}$$

2.5.9

$$\mathbf{E} = \frac{\mathbf{F}}{Q_t} \qquad \text{(definition)}$$

2.5.10

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R^2}\mathbf{a}_R$$

2.5.11

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho_L\,dl}{R^2}\mathbf{a}_R = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho_S\,dS}{R^2}\mathbf{a}_R = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho_v\,dv}{R^2}\mathbf{a}_R$$

2.5.12

$$E = 9\times 10^9\frac{1\times 10^{-9}}{(0.1)^2} = 9\times 10^9\times 10^{-7} = 900\,\mathrm{V/m}$$

2.6 Electric Dipole & Polarization

2.6.1

$$V = \frac{Qd\cos\theta}{4\pi\varepsilon_0 r^2} = \frac{p\cos\theta}{4\pi\varepsilon_0 r^2} = \frac{\mathbf{p}\cdot\hat{a}_r}{4\pi\varepsilon_0 r^2} \qquad (r \gg d)$$

2.6.2

$$\mathbf{E} = \frac{p}{4\pi\varepsilon_0 r^3}\big(2\cos\theta\,\hat{a}_r + \sin\theta\,\hat{a}_\theta\big), \qquad |\mathbf{E}| = \frac{p}{4\pi\varepsilon_0 r^3}\sqrt{1+3\cos^2\theta}$$

2.6.3

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0 r^3}\big[3(\mathbf{p}\cdot\hat{a}_r)\hat{a}_r - \mathbf{p}\big]$$

2.6.4

$$\mathbf{P} = \lim_{\Delta v\to 0}\frac{\sum_k \mathbf{p}_k}{\Delta v} \qquad \text{(dipole moment per unit volume, C/m}^2\text{)}$$

2.6.5

$$\mathbf{P} = \varepsilon_0\chi_e\mathbf{E}, \qquad \mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P} = \varepsilon_0(1+\chi_e)\mathbf{E} = \varepsilon_0\varepsilon_r\mathbf{E}$$

2.6.6

$$\varepsilon_r = 1+\chi_e \quad \Longrightarrow \quad \mathbf{P} = \varepsilon_0(\varepsilon_r - 1)\mathbf{E} = \mathbf{D}\Big(1 - \frac{1}{\varepsilon_r}\Big)$$

2.6.7

$$E_z \approx \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3} = \frac{1}{4\pi\varepsilon_0}\frac{2Qd}{r^3}$$

2.6.8

$$\mathbf{P} = \varepsilon_0\chi_e\mathbf{E}, \qquad \mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P} = \varepsilon\mathbf{E}$$

2.6.9

$$E_{1t} = E_{2t}, \qquad D_{2n} - D_{1n} = \rho_S, \qquad V_1 = V_2$$

2.6.10

$$p = Qd = (1\times 10^{-9})(0.01) = 10^{-11}\,\mathrm{C\cdot m}$$

2.6.11

$$E = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3} = 9\times 10^9\frac{2\times 10^{-11}}{(0.5)^3} = 9\times 10^9\frac{2\times 10^{-11}}{0.125} = 1.44\,\mathrm{V/m}$$

2.7 Electric Potential & Energy Density

(K) Potential of standard configurations

Source	Potential	Reference
Point charge Q	$V = \dfrac{Q}{4\pi\varepsilon_0 R}$	$V = 0$ at $R = \infty$
Line charge ρ_L	$V_A - V_B = \dfrac{\rho_L}{2\pi\varepsilon_0} \ln \dfrac{\rho_B}{\rho_A}$	$V = 0$ at ∞ is impossible; use a finite reference
Sheet charge ρ_s	$V_A - V_B = -\dfrac{\rho_s}{2\varepsilon_0}(x_A - x_B)$	sheet at $x = 0$, valid for $x > 0$

2.7.1

$$V_{AB} = V_A - V_B = -\int_B^A \mathbf{E} \cdot d\mathbf{L}, \qquad W = Q(V_B - V_A)$$

2.7.2

$$\mathbf{E} = -\nabla V, \qquad \rho_v = \nabla \cdot \mathbf{D} = -\varepsilon \nabla^2 V$$

2.7.3

$$\nabla V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z$$

2.7.4

$$\nabla V = \frac{\partial V}{\partial r}\hat{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\hat{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\hat{a}_\phi$$

2.7.5

$$W_E = \frac{1}{2}\sum_{k=1}^{n} Q_k V_k = \frac{1}{2}\int_v \rho_v V\, dv = \frac{1}{2}\int_v \mathbf{D} \cdot \mathbf{E}\, dv$$

2.7.6

$$w_E = \frac{dW_E}{dv} = \frac{1}{2}\mathbf{D} \cdot \mathbf{E} = \frac{1}{2}\varepsilon E^2 = \frac{D^2}{2\varepsilon} \ \text{J/m}^3$$

2.7.7

$$\mathbf{D} = \varepsilon_0 \mathbf{E} \quad \text{(free space)}, \qquad \mathbf{D} = \varepsilon \mathbf{E} = \varepsilon_r \varepsilon_0 \mathbf{E} \quad \text{(linear dielectric)}$$

2.7.8

$$\Psi = \int_S \mathbf{D} \cdot d\mathbf{S}$$

2.7.9

$$\mathbf{D} = \varepsilon_0 \mathbf{E} = (8.854 \times 10^{-12})(100)\,\mathbf{a}_x = 8.854 \times 10^{-10}\,\mathbf{a}_x\,\text{C/m}^2$$

2.8 Poisson's & Laplace's Equations, Capacitance & Uniqueness Theorem

(Y) The three standard results (derived below)

Capacitor	Geometry	Capacitance
Parallel plate	area S, separation d	$C = \varepsilon S/d$
Coaxial cable	radii $a < b$, length L	$C = 2\pi\varepsilon L/\ln(b/a)$
Spherical	radii $a < b$	$C = 4\pi\varepsilon/\big(1/a - 1/b\big) = 4\pi\varepsilon ab/(b-a)$

2.8.1

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} \quad \text{(Poisson)}, \qquad \nabla^2 V = 0 \quad \text{(Laplace, charge-free region)}$$

2.8.2

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

2.8.3

$$\nabla^2 V = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

2.8.4

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 V}{\partial \phi^2}$$

2.8.5

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} \quad \text{(Poisson)}, \qquad \nabla^2 V = 0 \quad \text{(Laplace)}$$

2.8.6

$$\frac{d^2 V}{dx^2} = 0 \implies V = Ax + B, \qquad \mathbf{E} = -\frac{dV}{dx}\mathbf{a}_x$$

2.8.7

$$V = Ax + B; \quad V(0) = 0 \Rightarrow B = 0; \quad V(0.02) = 100 \Rightarrow A = 5000$$

2.8.8

$$V = 5000x \qquad (x \text{ in metres, } V \text{ in volts})$$

2.8.9

$$\mathbf{E} = -\nabla V = -5000\,\mathbf{a}_x\ \text{V/m} = -5\,\text{kV/m}\,\mathbf{a}_x$$

Chapter 3: Magnetic Field

5 topics · 37 items

3.1 Reference Formulas for Chapter 3 *(Reference)*

3.1.1

$$\mathrm{d}\mathbf{H} = \frac{I\,\mathrm{d}\mathbf{L} \times \hat{a}_R}{4\pi R^2} = \frac{I\,\mathrm{d}\mathbf{L} \times \mathbf{R}}{4\pi R^3} \quad \text{(A/m)}$$

3.1.2

$$\mathbf{H} = \frac{I}{4\pi\rho}\left(\sin\alpha_2 - \sin\alpha_1\right)\hat{a}_\phi$$

3.1.3

$$\mathbf{H} = \frac{I}{2\pi\rho}\,\hat{a}_\phi$$

3.1.4

$$\oint_L \mathbf{H}\cdot \mathrm{d}\mathbf{L} = I_{\text{enc}} \qquad \text{(integral form)}, \qquad \nabla\times\mathbf{H} = \mathbf{J} \qquad \text{(point form)}$$

3.1.5

$$\nabla\times\mathbf{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right)\hat{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}\right)\hat{a}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right)\hat{a}_z$$

3.1.6

$$\nabla\times\mathbf{H} = \left(\frac{1}{\rho}\frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z}\right)\hat{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho}\right)\hat{a}_\phi + \frac{1}{\rho}\left(\frac{\partial(\rho H_\phi)}{\partial \rho} - \frac{\partial H_\rho}{\partial \phi}\right)\hat{a}_z$$

3.1.7

$$\mathbf{F} = I\,\mathrm{d}\mathbf{L}\times\mathbf{B}, \qquad \mathbf{B} = \mu\mathbf{H}, \qquad \mu_0 = 4\pi\times10^{-7}\ \text{H/m}$$

3.1.8

$$\hat{a}_x\times\hat{a}_y = \hat{a}_z, \qquad \hat{a}_y\times\hat{a}_z = \hat{a}_x, \qquad \hat{a}_z\times\hat{a}_x = \hat{a}_y$$

3.1.9

$$\hat{a}_y \times \hat{a}_x = -\hat{a}_z, \qquad \hat{a}_z \times \hat{a}_y = -\hat{a}_x, \qquad \hat{a}_x \times \hat{a}_z = -\hat{a}_y$$

3.1.10

$$\hat{a}_\phi = -\sin\phi\,\hat{a}_x + \cos\phi\,\hat{a}_y, \qquad \cos\phi = \frac{x}{\rho}, \quad \sin\phi = \frac{y}{\rho}$$

3.2 Biot–Savart Law & Ampere's Circuital Law

3.2.1

$$d\mathbf{H} = \frac{1}{4\pi}\frac{I\,d\mathbf{L} \times \mathbf{a}_R}{R^2}$$

3.2.2

$$H = \frac{I}{2a} \qquad (z = 0 \text{ plane, loop centre})$$

3.2.3

$$H = \frac{I}{4\pi\rho}(\sin\alpha_1 + \sin\alpha_2)$$

3.2.4

$$H = \frac{I}{2a} = \frac{10}{2 \times 0.1} = 50\,\mathrm{A/m}$$

3.3 Curl & Stokes's Theorem

(K) Differential surface elements — needed for the right-hand side

Surface	$\mathrm{d}\mathbf{S}$	Boundary element $\mathrm{d}\mathbf{L}$
Plane $z = \text{const}$	$\mathrm{d}x\,\mathrm{d}y\,\hat{a}_z$	$\mathrm{d}x\hat{a}_x$ or $\mathrm{d}y\hat{a}_y$
Cylinder $\rho = \text{const}$	$\rho\,\mathrm{d}\phi\,\mathrm{d}z\,\hat{a}_\rho$	$\rho\,\mathrm{d}\phi\hat{a}_\phi$ or $\mathrm{d}z\hat{a}_z$
Sphere $r = \text{const}$	$r^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi\,\hat{a}_r$	$r\,\mathrm{d}\theta\hat{a}_\theta$ or $r\sin\theta\,\mathrm{d}\phi\hat{a}_\phi$
Cone $\theta = \text{const}$	$r\sin\theta\,\mathrm{d}r\,\mathrm{d}\phi\,\hat{a}_\theta$	$\mathrm{d}r\hat{a}_r$ or $r\sin\theta\,\mathrm{d}\phi\hat{a}_\phi$

3.3.1

$$(\nabla \times \mathbf{H}) \cdot \hat{a}_N = \lim_{\Delta S_N \to 0} \frac{\oint_L \mathbf{H} \cdot \mathrm{d}\mathbf{L}}{\Delta S_N}$$

3.3.2

$$\oint_L \mathbf{H} \cdot \mathrm{d}\mathbf{L} = \int_S (\nabla \times \mathbf{H}) \cdot \mathrm{d}\mathbf{S}$$

3.3.3

$$(\nabla \times \mathbf{A})_r = \frac{1}{r\sin\theta}\left(\frac{\partial(A_\phi \sin\theta)}{\partial\theta} - \frac{\partial A_\theta}{\partial\phi}\right)$$

3.3.4

$$(\nabla \times \mathbf{A})_\theta = \frac{1}{r\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{1}{r}\frac{\partial(rA_\phi)}{\partial r}$$

3.3.5

$$(\nabla \times \mathbf{A})_\phi = \frac{1}{r}\left(\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right)$$

3.3.6

$$\nabla \times \mathbf{A} = \lim_{\Delta S \to 0} \frac{1}{\Delta S}\oint_L \mathbf{A} \cdot d\mathbf{L}\,\mathbf{a}_n$$

3.3.7

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_L \mathbf{A} \cdot d\mathbf{L}$$

3.3.8

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -y & x & 0 \end{vmatrix} = \mathbf{a}_z \left(\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(-y) \right) = 2\mathbf{a}_z$$

3.4 Magnetic Force, Torque & Boundary Conditions

3.4.1

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \qquad \mathbf{F} = I \int d\mathbf{L} \times \mathbf{B}$$

3.4.2

$$\mathbf{m} = I\mathbf{S}, \qquad \mathbf{T} = \mathbf{m} \times \mathbf{B}$$

3.4.3

$$F = ILB = (5)(2)(0.5) = 5\,\mathrm{N}$$

3.5 Scalar & Vector Magnetic Potential

(N) Magnetic boundary conditions

	Continuous tangential	Continuous normal
Electric	$E_{t1} = E_{t2}$ always	$D_{N1} = D_{N2}$ if $\rho_s = 0$
Magnetic	$H_{t1} = H_{t2}$ only if $\mathbf{K} = \mathbf{0}$	$B_{N1} = B_{N2}$ always

3.5.1

$$\mathbf{H} = -\nabla V_m \qquad \text{(valid ONLY where } \mathbf{J} = \mathbf{0}\text{)}$$

3.5.2

$$\nabla^2 V_m = 0 \qquad \text{(units of } V_m\text{: amperes)}$$

3.5.3

$$\mathbf{B} = \nabla \times \mathbf{A} \qquad \text{(valid EVERYWHERE)}$$

3.5.4

$$\mathbf{A} = \frac{\mu I \, \mathrm{d}\mathbf{L}}{4\pi R} \quad \text{(filament)}, \qquad \nabla^2 \mathbf{A} = -\mu \mathbf{J}, \qquad \Phi = \int_S \mathbf{B} \cdot \mathrm{d}\mathbf{S} = \oint_L \mathbf{A} \cdot \mathrm{d}\mathbf{L}$$

3.5.5

$$B_{N1} = B_{N2} \implies \mu_1 H_{N1} = \mu_2 H_{N2}$$

3.5.6

$$\big(\mathbf{H}_1 - \mathbf{H}_2\big) \times \hat{a}_{N12} = \mathbf{K} \implies H_{t1} = H_{t2} \text{ when } \mathbf{K} = \mathbf{0}$$

3.5.7

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_1}{\mu_2} \qquad (\theta \text{ measured from the normal})$$

3.5.8

$$\mathbf{B} = \nabla \times \mathbf{A}$$

3.5.9

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{I\, d\mathbf{L}}{R} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}}{R}\, dv$$

3.5.10

$$\mathbf{B} = \nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \mu_0 y & 0 & 0 \end{vmatrix} = -\mu_0 \mathbf{a}_z$$

Chapter 4: Time-Varying Fields

4 topics · 17 items

4.1 Reference Formulas for Chapter 4 *(Reference)*

(B) The three ways flux can change

Case	Name	EMF
Loop fixed, **B** varies	transformer emf	$\mathcal{E} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$
Loop moves, **B** static	motional emf	$\mathcal{E} = \oint_L (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$
Both	general case	$\mathcal{E} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} + \oint_L (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$

(C) Maxwell's equations — time-varying fields

Law	Point form	Integral form
Faraday	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_L \mathbf{E} \cdot d\mathbf{L} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S}$
Ampere (corrected)	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_L \mathbf{H} \cdot d\mathbf{L} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}$
Gauss (electric)	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}$
Gauss (magnetic)	$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$

4.1.1

$$\mathcal{E} = -\frac{d\Phi}{dt} \quad \text{(volts)}, \qquad \mathcal{E} = -N\frac{d\Phi}{dt} \quad (N \text{ turns}), \qquad \oint_L \mathbf{E} \cdot d\mathbf{L} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S}$$

4.1.2

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S}.$$

4.1.3

$$\nabla \times \mathbf{E}_s = -j\omega\mu\mathbf{H}_s, \qquad \nabla \times \mathbf{H}_s = (\sigma + j\omega\varepsilon)\mathbf{E}_s, \qquad \nabla \cdot \mathbf{E}_s = \frac{\rho_{vs}}{\varepsilon}, \qquad \nabla \cdot \mathbf{H}_s = 0.$$

4.1.4

$$\mathbf{J}_c = \sigma\mathbf{E} \quad \text{(conduction)}, \qquad \mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad \text{(displacement)}.$$

4.1.5

$$\frac{|\mathbf{J}_c|}{|\mathbf{J}_d|} = \frac{\sigma}{\omega\varepsilon} = \frac{\sigma}{2\pi f\varepsilon} = \tan\delta.$$

4.2 Displacement Current

4.2.1

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} = \varepsilon \frac{\partial \mathbf{E}}{\partial t}$$

4.2.2

$$I_d = \int_S \mathbf{J}_d \cdot d\mathbf{S} = \frac{d\Psi}{dt} = C\frac{dV}{dt} \quad \text{(parallel-plate capacitor)}$$

4.2.3

$$i = C\frac{dv}{dt} = (200 \times 10^{-12})(1000)(1000)\cos(1000t) = 0.2 \times 10^{-3}\cos(1000t)\,\text{A}$$

4.3 Faraday's Law, Motional & Transformer EMF

4.3.1

$$\text{EMF} = \oint \mathbf{E} \cdot d\mathbf{L} = -\frac{d\Phi}{dt} = -\frac{d}{dt}\int_S \mathbf{B} \cdot d\mathbf{S}$$

4.3.2

$$e = -\frac{d\Phi}{dt} = -0.5(-100\sin(100t)) = 50\sin(100t)\,\text{V}$$

4.4 Maxwell's Equations

4.4.1

$$\nabla \cdot \mathbf{D} = \rho_v, \qquad \nabla \cdot \mathbf{B} = 0$$

4.4.2

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

4.4.3

$$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}, \qquad \oint \mathbf{B} \cdot d\mathbf{S} = 0$$

4.4.4

$$\oint \mathbf{E} \cdot d\mathbf{L} = -\frac{d}{dt}\int \mathbf{B} \cdot d\mathbf{S}, \qquad \oint \mathbf{H} \cdot d\mathbf{L} = I_{\text{enc}} + \frac{d}{dt}\int \mathbf{D} \cdot d\mathbf{S}$$

4.4.5

$$H = \frac{E}{\eta_0} = \frac{10}{377} \approx 0.0265\,\text{A/m}$$

Chapter 5: Plane Waves

5 topics · 38 items

5.1 Reference Formulas for Chapter 5 *(Reference)*

(E) The three standard media

Quantity	Free space ($\sigma = 0$)	Lossless dielectric ($\sigma = 0$)	Good conductor ($\sigma \gg \omega\varepsilon$)
α	0	0	$\sqrt{\pi f \mu \sigma}$
β	$\omega\sqrt{\mu_0\varepsilon_0}$	$\omega\sqrt{\mu\varepsilon}$	$\sqrt{\pi f \mu \sigma}$
η	$377\,\Omega$	$\sqrt{\mu/\varepsilon}$	$\sqrt{\omega\mu/\sigma}\,\angle 45^\circ$
u	3×10^8 m/s	$1/\sqrt{\mu\varepsilon}$	$2\omega/\mu\sigma$ (equiv. $\omega\delta$)

(O) Normal incidence

Case	Γ	τ	Meaning
$\eta_2 = \eta_1$ (matched)	0	1	no reflection
$\eta_2 = 0$ (PEC)	-1	0	total reflection, phase reversed
$\eta_2 = \infty$	$+1$	2	total reflection, no reversal

5.1.1

$$\nabla^2 \mathbf{E} = \mu\varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \text{(lossless, time domain)}$$

5.1.2

$$\nabla^2 \mathbf{E}_s = \gamma^2 \mathbf{E}_s, \qquad \nabla^2 \mathbf{H}_s = \gamma^2 \mathbf{H}_s \quad \text{(Helmholtz, phasor form)}$$

5.1.3

$$\gamma = \alpha + j\beta = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$$

5.1.4

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right)} \quad \text{(Np/m)}$$

5.1.5

$$\beta = \omega\sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1\right)} \quad \text{(rad/m)}$$

5.1.6

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = |\eta|\angle\theta_\eta\ \Omega$$

5.1.7

$$u = \frac{\omega}{\beta}\ \text{(m/s)}, \qquad \lambda = \frac{2\pi}{\beta} = \frac{u}{f}\ \text{(m)}, \qquad \delta = \frac{1}{\alpha}\ \text{(skin depth, m)}$$

5.1.8

$$\tan\delta = \frac{\sigma}{\omega\varepsilon}\begin{cases} \ll 1 & \text{good dielectric} \\ \gg 1 & \text{good conductor} \end{cases}$$

5.1.9

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi \approx 377\,\Omega, \qquad c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 3\times 10^8\ \text{m/s}$$

5.1.10

$$\nabla\times\nabla\times\mathbf{A} = \nabla(\nabla\cdot\mathbf{A}) - \nabla^2\mathbf{A}$$

5.1.11

$$\mathbf{P} = \mathbf{E}\times\mathbf{H}\ \text{W/m}^2 \quad \text{(instantaneous)}$$

5.1.12

$$\mathbf{P}_{\text{avg}} = \frac{1}{2}\,\text{Re}\{\mathbf{E}_s\times\mathbf{H}_s^*\} \quad \text{(time-average, phasor)}$$

5.1.13

$$\oint_S(\mathbf{E}\times\mathbf{H})\cdot d\mathbf{S} = -\frac{\partial}{\partial t}\int_v\left(\frac{1}{2}\varepsilon E^2 + \frac{1}{2}\mu H^2\right)dv - \int_v \sigma E^2\,dv$$

5.1.14

$$\mathbf{P}_{\text{avg}} = \frac{E_0^2}{2|\eta|}e^{-2\alpha z}\cos\theta_\eta\,\hat{\mathbf{a}}_z \quad \text{(lossy)}$$

5.1.15

$$\mathbf{P}_{\text{avg}} = \frac{E_0^2}{2\eta}\,\hat{\mathbf{a}}_z \quad (\text{lossless: } \alpha = 0,\ \theta_\eta = 0)$$

5.1.16

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f\mu\sigma}}, \qquad \alpha = \beta = \sqrt{\pi f\mu\sigma}, \qquad \gamma = (1+j)\sqrt{\pi f\mu\sigma}$$

5.1.17

$$\eta = \sqrt{\frac{\omega\mu}{\sigma}}\angle 45^\circ, \qquad u = \frac{\omega}{\beta} = \omega\delta, \qquad \lambda = 2\pi\delta$$

5.1.18

$$\alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}, \qquad \beta \approx \omega\sqrt{\mu\varepsilon}, \qquad \eta \approx \sqrt{\frac{\mu}{\varepsilon}}$$

5.1.19

$$\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \qquad \tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1}, \qquad 1 + \Gamma = \tau$$

5.1.20

$$s = \frac{|E|_{\max}}{|E|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \qquad |\Gamma| = \frac{s-1}{s+1}, \qquad 1 \le s \le \infty$$

5.1.21

$$\eta_{\text{in}} = \eta_2 \frac{\eta_3 \cos\beta_2 d + j\eta_2 \sin\beta_2 d}{\eta_2 \cos\beta_2 d + j\eta_3 \sin\beta_2 d}$$

5.1.22

$$d = \frac{\lambda_2}{4} : \ \eta_{\text{in}} = \frac{\eta_2^2}{\eta_3} \quad \text{(quarter-wave);} \qquad d = \frac{\lambda_2}{2} : \ \eta_{\text{in}} = \eta_3 \quad \text{(half-wave window)}$$

5.2 Numerical Problems on Uniform Plane Waves

5.2.1

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} = \frac{1}{\alpha}, \qquad \alpha = \beta = \frac{1}{\delta}, \qquad \eta \approx (1+j)\frac{1}{\sigma\delta}$$

5.2.2

$$\delta = \sqrt{\frac{2}{\omega\mu_0\sigma}} = \sqrt{\frac{2}{2\pi \times 10^6\,(4\pi \times 10^{-7})\,(5.8 \times 10^7)}} \approx 66.1\,\mu\text{m}$$

5.3 Poynting Theorem, Skin Depth & Loss Tangent

5.3.1

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

5.3.2

$$\mathbf{P}_{\text{av}} = \frac{1}{2}\text{Re}\{\mathbf{E}_s \times \mathbf{H}_s^*\}$$

5.3.3

$$P_{\text{av}} = \frac{E^2}{2\eta}$$

5.3.4

$$P_{\text{av}} = \frac{E^2}{2\eta_0} = \frac{100}{2 \times 377} \approx 0.133\,\text{W/m}^2$$

5.4 Reflection of Plane Waves & Standing Wave Ratio

5.4.1

$$\Gamma = \frac{E_r}{E_i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}, \qquad \tau = \frac{E_t}{E_i} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

5.4.2

$$\tau = 1 + \Gamma$$

5.4.3

$$\Gamma = \frac{188.5 - 377}{188.5 + 377} = \frac{-188.5}{565.5} \approx -0.431$$

5.4.4

$$\tau = \frac{2 \times 188.5}{565.5} \approx 0.569$$

5.5 Wave Propagation & Field Derivations

5.5.1

$$\gamma = \alpha + j\beta = j\omega\sqrt{\mu\varepsilon}\sqrt{1 - \frac{j\sigma}{\omega\varepsilon}}$$

5.5.2

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

5.5.3

$$\omega = 2\pi \times 10^8, \qquad \beta = \omega\sqrt{\mu_0\varepsilon_0\varepsilon_r} = \frac{\omega}{c}\sqrt{\varepsilon_r} = \frac{2\pi \times 10^8}{3 \times 10^8} \times 2 = 4.189\,\mathrm{rad/m}$$

5.5.4

$$v_p = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{2} = 1.5 \times 10^8\,\mathrm{m/s}$$

Chapter 6: Transmission Lines

3 topics · 25 items

6.1 Reference Formulas for Chapter 6 *(Reference)*

(D) The two special lines

	Lossless ($R = G = 0$)	Distortionless ($R/L = G/C$)
α	0	$\sqrt{RG}$ (independent of ω)
β	$\omega\sqrt{LC}$	$\omega\sqrt{LC}$
γ	$j\omega\sqrt{LC}$	$\sqrt{RG} + j\omega\sqrt{LC}$
Z_0	$\sqrt{L/C}$ (real)	$\sqrt{L/C}$ (real)
u	$1/\sqrt{LC}$	$1/\sqrt{LC}$

(G) Special cases worth knowing by sight

Condition	Z_{in}	Name
$\ell = \lambda/2$ (or any $n\lambda/2$)	Z_L	half-wave repeater
$\ell = \lambda/4$	Z_0^2/Z_L	quarter-wave transformer
$Z_L = Z_0$	Z_0	matched — no reflection
$Z_L = 0$ (short)	$jZ_0 \tan\beta\ell$	purely reactive
$Z_L = \infty$ (open)	$-jZ_0 \cot\beta\ell$	purely reactive

6.1.1

$$R\ (\Omega/\mathrm{m}), \quad L\ (\mathrm{H/m}), \quad G\ (\mathrm{S/m}), \quad C\ (\mathrm{F/m})$$

6.1.2

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

6.1.3

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

6.1.4

$$u = \frac{\omega}{\beta}, \qquad \lambda = \frac{2\pi}{\beta}$$

6.1.5

$$\gamma Z_0 = R + j\omega L, \qquad \frac{\gamma}{Z_0} = G + j\omega C$$

6.1.6

$$Z_{\text{in}} = Z_0 \frac{Z_L + jZ_0 \tan\beta\ell}{Z_0 + jZ_L \tan\beta\ell} \qquad \text{(lossless)}$$

6.1.7

$$Z_{\text{in}} = Z_0 \frac{Z_L + Z_0 \tanh\gamma\ell}{Z_0 + Z_L \tanh\gamma\ell} \qquad \text{(lossy)}$$

6.1.8

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}, \qquad s = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

6.2 Input Impedance, Reflection Coefficient & SWR

(J) Power

Condition	Z_{in}	Reason
$Z_l = Z_0$	Z_0	numerator and denominator share the factor
$\ell = n\lambda/2$	Z_l	$\tan\beta\ell = 0$
$\ell = \lambda/4$	Z_0^2/Z_l	$\tan\beta\ell \to \infty$
$Z_l = 0$ (short)	$jZ_0 \tan\beta\ell$	purely reactive
$Z_l = \infty$ (open)	$-jZ_0 \cot\beta\ell$	purely reactive
$\ell \to \infty$ (lossy)	Z_0	$\tanh\gamma\ell \to 1$

6.2.1

$$V(d) = V_0^+\left(e^{j\beta d} + \Gamma_L e^{-j\beta d}\right)$$

6.2.2

$$V_L = V_0^+(1 + \Gamma_L) \qquad (\text{at } d = 0)$$

6.2.3

$$|V|_{\max} = |V_0^+|(1 + |\Gamma_L|), \qquad |V|_{\min} = |V_0^+|(1 - |\Gamma_L|), \qquad s = \frac{|V|_{\max}}{|V|_{\min}}$$

6.2.4

$$d_{\max} = \frac{\phi}{2\beta} = \frac{\phi_{\text{rad}}\,\lambda}{4\pi}$$

6.2.5

$$V_{\text{in}} = V_g \frac{Z_{\text{in}}}{Z_g + Z_{\text{in}}}, \qquad I_{\text{in}} = \frac{V_g}{Z_g + Z_{\text{in}}}$$

6.2.6

$$P = \frac{1}{2}\text{Re}\{VI^*\} = \frac{1}{2}|V|^2\,\text{Re}\left\{\frac{1}{Z^*}\right\} = \frac{1}{2}|I|^2\,\text{Re}\{Z\}$$

6.2.7

$$P_L = \frac{|V_L|^2}{2R_L} \qquad \text{(resistive load)}$$

6.2.8

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}, \qquad \text{SWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

6.2.9

$$Z_{\text{in}} = Z_0 \frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$$

6.2.10

$$\Gamma = \frac{75 - 50}{75 + 50} = \frac{25}{125} = 0.2$$

6.2.11

$$\text{SWR} = \frac{1+0.2}{1-0.2} = \frac{1.2}{0.8} = 1.5$$

6.2.12

$$Z_{\text{in}} = \frac{Z_0^2}{Z_L} = \frac{50^2}{75} = \frac{2500}{75} = 33.33\,\Omega$$

6.3 Line Parameters (Lossless, Lossy & Distortionless)

6.3.1

$$\alpha = \sqrt{RG}, \qquad \beta = \omega\sqrt{LC}, \qquad Z_0 = \sqrt{L/C}$$

6.3.2

$$\frac{R}{L} = \frac{G}{C} \implies G = \frac{RC}{L} = \frac{20 \times 20 \times 10^{-9}}{0.5 \times 10^{-3}} = 4 \times 10^{-5}\,\text{S/m}$$